

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
BOIN12	17.5	12	4.8	1.4	0.9	63.4
U-BOIN	15.6	17.4	20.6	18.7	13.7	14
EffTox	0	0	3	1	2	95
AIDE one-stage (U2)	16.4	11.5	2.3	0.6	0.2	69
AIDE one-stage (U3)	16.7	11.4	1.8	0.4	0.1	69.6
AIDE two-stage (U2)	15.9	8	1.6	0.5	0.2	73.8
AIDE two-stage (U3)	17.3	7.7	1.3	0.3	0.2	73.2
Mean patients treated						
BOIN12	10	5.8	3.3	1.9	0.8	
U-BOIN	9	5.4	4.7	3.9	3.7	
EffTox	5.3	3.2	3	2.2	2	
AIDE one-stage (U2)	10.7	5.7	2.5	1.4	0.4	
AIDE one-stage (U3)	10.8	5.6	2.5	1.4	0.4	
AIDE two-stage (U2)	11.1	3.6	2.1	0.9	0.3	
AIDE two-stage (U3)	11.1	3.6	2.2	0.9	0.3	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
BOIN12	27.8	24.6	32.7	9.9	1.3	3.7
U-BOIN	11.4	13.5	41.5	21.8	10.6	1.2
EffTox	51	10	9	14	11	4
AIDE one-stage (U2)	23	17.3	39.6	15.4	2.1	2.6
AIDE one-stage (U3)	28.1	19	35.2	13.1	1.7	2.9
AIDE two-stage (U2)	31.7	15	33.2	11.2	3.7	5.2
AIDE two-stage (U3)	34.3	16.3	31.2	10.2	3	5
Mean patients treated						
BOIN12	8.5	7.6	8.6	3.6	1.1	
U-BOIN	8.8	7.2	8.2	4	1.6	
EffTox	35.5	6.2	4.7	5.2	6.7	
AIDE one-stage (U2)	9	6.9	8.9	3.9	0.9	
AIDE one-stage (U3)	9.3	7.1	8.8	3.6	0.7	
AIDE two-stage (U2)	10.8	5.1	7.2	3.8	2.2	
AIDE two-stage (U3)	10.8	5.1	7.1	3.9	2.1	

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda_T = 0.3; U3: Type = 3, Lambda_T = 1.

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
BOIN12	10.1	54	21.9	7.1	1.6	5.3
U-BOIN	2.4	45	23.6	17.1	10.2	1.7
EffTox	24	29	12	13	16	6
AIDE one-stage (U2)	7.9	44.5	24.9	13.2	2.4	7.1
AIDE one-stage (U3)	9.6	49.1	23.1	9.2	1.7	7.3
AIDE two-stage (U2)	10.5	39.5	21	8.2	2.7	18.1
AIDE two-stage (U3)	12.6	42.7	18.9	5.7	1.9	18.2
Mean patients treated						
BOIN12	6.4	11.4	6.4	3.5	1.4	
U-BOIN	7.2	10.6	6.5	3.7	1.7	
EffTox	21.4	13.7	7.9	5.6	8.7	
AIDE one-stage (U2)	6.1	11.1	6.8	3.8	1.1	
AIDE one-stage (U3)	6.3	11.8	6.5	3.4	0.9	
AIDE two-stage (U2)	7.8	7.5	6.1	3.6	2	
AIDE two-stage (U3)	7.8	7.6	6	3.6	2	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
BOIN12	12.6	14.6	26.9	35.6	7.1	3.2
U-BOIN	3.8	4	18.8	46.4	27	0
EffTox	0	3	30	29	33	4
AIDE one-stage (U2)	5.7	5.2	16.9	53.3	17.6	1.3
AIDE one-stage (U3)	8.6	7.7	16.4	51.3	14.9	1.1
AIDE two-stage (U2)	6.4	7.2	21.6	42.5	19.4	2.9
AIDE two-stage (U3)	10.6	8.8	18.7	41.8	17.2	2.9
Mean patients treated						
BOIN12	5	5.7	7.2	8.8	3.1	
U-BOIN	4.9	6.1	7.6	7.8	3.8	
EffTox	6.2	5.8	15.5	12.3	18.2	
AIDE one-stage (U2)	4.1	4.5	6.7	10.6	3.9	
AIDE one-stage (U3)	4.3	4.6	6.7	10.7	3.6	
AIDE two-stage (U2)	4.3	4.2	6.5	8	6.5	
AIDE two-stage (U3)	4.3	4.1	6.4	8	6.5	

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AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda_T = 0.3; U3: Type = 3, Lambda_T = 1.

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Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
BOIN12	5.9	44.9	28.6	13.4	5.9	1.3
U-BOIN	2.4	34.9	27.9	21.6	13.1	0.1
EffTox	6	25	24	10	26	9
AIDE one-stage (U2)	5.4	37.6	25.7	19.5	10.4	1.4
AIDE one-stage (U3)	9	42.6	23.2	17.5	6.2	1.5
AIDE two-stage (U2)	3.7	36.3	24.8	18.5	11.2	5.5
AIDE two-stage (U3)	6.9	42.5	22.5	16.3	6.3	5.5
Mean patients treated						
BOIN12	4.5	8.3	7.1	6	3.9	
U-BOIN	5.3	8.5	7	5.6	3.8	
EffTox	11.1	12.4	13.1	6.8	13	
AIDE one-stage (U2)	3.8	7.2	7	7.1	4.6	
AIDE one-stage (U3)	3.9	7.9	7.2	6.8	4	
AIDE two-stage (U2)	4.5	6.5	6.5	5.8	5.6	
AIDE two-stage (U3)	4.5	6.5	6.6	5.7	5.6	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
BOIN12	59	17	5.4	1.8	0.1	16.7
U-BOIN	21.5	17.1	15.8	19.3	15.1	11.2
EffTox	75	2	3	2	1	17
AIDE one-stage (U2)	61.9	17.6	5.1	1.9	0.5	13
AIDE one-stage (U3)	62.7	17.8	4.2	1.5	0.4	13.4
AIDE two-stage (U2)	66.7	13	3.6	1.3	0.4	15
AIDE two-stage (U3)	67.3	13.2	3.3	1	0.4	14.8
Mean patients treated						
BOIN12	15.3	6.5	3.3	1.6	0.6	
U-BOIN	11.7	6.3	4.4	3.1	1.9	
EffTox	45.3	2	1.6	1.3	1.3	
AIDE one-stage (U2)	17.1	6.3	2.7	1.3	0.3	
AIDE one-stage (U3)	17.6	6.2	2.4	1.2	0.3	
AIDE two-stage (U2)	19.2	4	2.5	1.1	0.4	
AIDE two-stage (U3)	19.1	4	2.6	1.1	0.4	

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda_T = 0.3; U3: Type = 3, Lambda_T = 1.

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
BOIN12	13.1	52.3	21.3	8.4	2.2	2.7
U-BOIN	5.4	48.3	26.2	13.2	5.2	1.7
EffTox	51	13	7	8	20	2
AIDE one-stage (U2)	11.8	46.8	22.7	12.5	3.5	2.7
AIDE one-stage (U3)	13.5	50.5	20.7	10.2	2.3	2.8
AIDE two-stage (U2)	22.1	37.1	20.3	11	3.6	5.9
AIDE two-stage (U3)	25.1	39.8	18.1	8.5	2.7	5.8
Mean patients treated						
BOIN12	6.8	12.8	5.9	2.9	1.1	
U-BOIN	7.9	10.8	6.5	3.2	1.3	
EffTox	36.4	7	4.8	3.3	7.5	
AIDE one-stage (U2)	7.4	12.5	5.8	3	0.8	
AIDE one-stage (U3)	7.7	13.2	5.4	2.6	0.7	
AIDE two-stage (U2)	10.1	7	5.5	3.8	2.5	
AIDE two-stage (U3)	10.1	7	5.6	3.8	2.5	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
BOIN12	29.9	15.1	4.2	1.3	0.2	49.3
U-BOIN	12.6	15.7	15.2	18	23.6	14.9
EffTox	4	7	11	6	2	70
AIDE one-stage (U2)	28.7	12.4	3.5	1.5	0.3	53.6
AIDE one-stage (U3)	28.2	14	2.4	1.3	0.2	53.9
AIDE two-stage (U2)	26.2	8.5	3.2	1	0.4	60.7
AIDE two-stage (U3)	27.4	8.8	2.3	0.9	0.3	60.3
Mean patients treated						
BOIN12	11.7	5.9	3.1	1.8	0.7	
U-BOIN	8.8	5	4.3	4.2	4.2	
EffTox	11.7	5	5.2	3.6	3.9	
AIDE one-stage (U2)	12.6	6	2.3	1.2	0.3	
AIDE one-stage (U3)	12.9	5.8	2.2	1.1	0.3	
AIDE two-stage (U2)	13.4	3.6	1.9	0.8	0.3	
AIDE two-stage (U3)	13.4	3.6	1.9	0.8	0.3	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
BOIN12	13.1	13.9	23.5	13	25.7	10.8
U-BOIN	3.7	5.4	12.6	20.2	57.7	0.4
EffTox	0	1	10	8	66	15
AIDE one-stage (U2)	10.4	11	21.2	17	27.6	12.8
AIDE one-stage (U3)	15.4	13.2	19.8	15.1	23.3	13.2
AIDE two-stage (U2)	7.7	8.6	17.8	13.6	26.4	25.9
AIDE two-stage (U3)	13.9	9.8	15.1	12.7	22.5	26
Mean patients treated						
BOIN12	5.7	5.7	5.9	5	6.4	
U-BOIN	6.6	5.7	6.3	5.4	6	
EffTox	5.9	4.2	7.8	5.8	28.9	
AIDE one-stage (U2)	6.2	5.6	6	5.4	5.2	
AIDE one-stage (U3)	6.4	5.7	5.9	5.4	4.8	
AIDE two-stage (U2)	6.9	4.1	5.3	4.1	5.2	
AIDE two-stage (U3)	6.8	4.1	5.3	4.1	5.3	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
BOIN12	8.8	42.9	16.2	5.4	1.5	25.2
U-BOIN	1	31.6	24.8	19.6	18.6	4.4
EffTox	6	24	14	17	25	14
AIDE one-stage (U2)	4.7	36.3	15.9	6.5	1.8	34.8
AIDE one-stage (U3)	5.2	38	15.1	5.5	1.1	35.1
AIDE two-stage (U2)	6.4	29.1	12	4.3	1.9	46.3
AIDE two-stage (U3)	6.9	30.5	10.7	4	1.6	46.3
Mean patients treated						
BOIN12	7.3	10.1	5.2	2.6	1	
U-BOIN	7.7	9	6.3	3.8	2.1	
EffTox	12.8	12.5	7.4	6.7	14.2	
AIDE one-stage (U2)	7.5	9.7	4.8	2.2	0.7	
AIDE one-stage (U3)	7.6	10.1	4.6	2.1	0.6	
AIDE two-stage (U2)	8.4	5.9	4.1	2.2	1.1	
AIDE two-stage (U3)	8.4	6	4.1	2.2	1.2	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
BOIN12	57.6	22.5	5.7	0.5	0	13.7
U-BOIN	42.3	22.3	12.8	8.3	9.8	4.5
EffTox	49	5	4	1	0	41
AIDE one-stage (U2)	57.3	23.4	4.4	0.6	0	14.3
AIDE one-stage (U3)	59.9	21.1	4.5	0.6	0	13.9
AIDE two-stage (U2)	58.1	20.8	4.6	1	0.1	15.4
AIDE two-stage (U3)	60.4	19.2	4.4	0.6	0	15.4
Mean patients treated						
BOIN12	14.4	7.8	3.9	1.6	0.5	
U-BOIN	11.6	6.7	4.7	3.3	2.7	
EffTox	33.5	4	2.8	1.4	0.7	
AIDE one-stage (U2)	15.4	7.8	3.4	1.4	0.2	
AIDE one-stage (U3)	15.9	7.6	3.2	1.3	0.2	
AIDE two-stage (U2)	16.8	5.9	3.5	1.1	0.2	
AIDE two-stage (U3)	16.9	5.7	3.6	1	0.2	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
BOIN12	16.6	23.8	15.1	32.3	9.9	2.3
U-BOIN	6.8	10.1	6.4	39.1	37.6	0
EffTox	0	1	20	43	28	8
AIDE one-stage (U2)	10.7	10.7	8.3	43.9	24.3	2.1
AIDE one-stage (U3)	14.5	13.6	7.1	42.4	20.4	2
AIDE two-stage (U2)	12.2	11.4	9.6	37.6	23.8	5.4
AIDE two-stage (U3)	18.8	11.5	7.7	34.7	21.8	5.5
Mean patients treated						
BOIN12	5.2	6.2	5.9	8.2	4.4	
U-BOIN	5.2	6.3	6.5	7.4	4.8	
EffTox	6	4.2	12.5	15.4	18.7	
AIDE one-stage (U2)	4.4	5	5.4	9.5	5.5	
AIDE one-stage (U3)	4.6	5.1	5.4	9.6	5.2	
AIDE two-stage (U2)	4.7	4.5	5.2	7.5	7.3	
AIDE two-stage (U3)	4.7	4.5	5.3	7.4	7.2	

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda_T = 0.3; U3: Type = 3, Lambda_T = 1.

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
BOIN12	12.2	41.9	14.7	4.6	2	24.6
U-BOIN	3.2	25.1	28.2	21	19.1	3.4
EffTox	10	20	15	14	18	23
AIDE one-stage (U2)	6.3	36.8	17.2	6.7	1.7	31.3
AIDE one-stage (U3)	8.5	37.6	15.9	5.1	1.3	31.6
AIDE two-stage (U2)	8.3	29.4	13.8	5.1	2	41.4
AIDE two-stage (U3)	9.5	30.2	12.9	4.6	1.3	41.5
Mean patients treated						
BOIN12	7.9	9.4	5.1	2.9	1.4	
U-BOIN	7.8	7.5	6.5	4.4	3	
EffTox	13.4	11	7.7	6.4	10.8	
AIDE one-stage (U2)	7.6	9.1	5	2.9	0.9	
AIDE one-stage (U3)	7.7	9.4	4.9	2.7	0.8	
AIDE two-stage (U2)	8.2	6.2	4.4	2.4	1.2	
AIDE two-stage (U3)	8.2	6.2	4.5	2.4	1.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
BOIN12	23.8	21.2	16.7	22.8	8.6	6.9
U-BOIN	11.4	8.4	10.1	35.8	34.1	0.2
EffTox	0	1	18	23	34	24
AIDE one-stage (U2)	14.9	15.1	14.4	34.2	16	5.4
AIDE one-stage (U3)	21.3	18.2	13.2	30.6	11.2	5.5
AIDE two-stage (U2)	16.2	11.7	13.7	29	16.6	12.8
AIDE two-stage (U3)	24.8	14.3	9.8	26.3	11.9	12.9
Mean patients treated						
BOIN12	6.2	6.7	6	6.6	4	
U-BOIN	6.2	6.3	6.1	6.6	5	
EffTox	5.6	4.3	11.3	9.9	19.2	
AIDE one-stage (U2)	5.7	5.9	5.7	7.8	4.3	
AIDE one-stage (U3)	6	6.1	5.6	7.8	4	
AIDE two-stage (U2)	5.8	5.1	5.7	6.2	5.2	
AIDE two-stage (U3)	5.9	5.1	5.7	6.2	5.2	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
BOIN12	42.4	26.6	12.6	3.7	0.9	13.8
U-BOIN	36.2	23.6	14.4	12.5	12.3	1
EffTox	3	2	4	3	4	83
AIDE one-stage (U2)	41.9	27.2	13.1	6.5	0.4	10.9
AIDE one-stage (U3)	46.1	28.2	10	3.3	0.5	11.9
AIDE two-stage (U2)	43.2	23.6	11.2	3.2	0.9	17.9
AIDE two-stage (U3)	52.3	20.1	6.9	2.3	0.4	18
Mean patients treated						
BOIN12	9.6	8.3	5.5	3.6	1.8	
U-BOIN	9.2	7.5	5.6	4.2	3.4	
EffTox	11.3	4.4	5.1	3.7	3.3	
AIDE one-stage (U2)	9.4	8	5.7	4.1	1.6	
AIDE one-stage (U3)	9.9	8	5.5	4	1.5	
AIDE two-stage (U2)	10	6.5	5.8	3.3	1.7	
AIDE two-stage (U3)	10	6.6	5.8	3.3	1.7	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
BOIN12	35.1	27.5	19.7	12.1	5.6	0
U-BOIN	28.1	26.8	21.1	16.3	7.7	0
EffTox	46	5	12	9	9	19
AIDE one-stage (U2)	40.6	24.4	14.6	13.1	7.3	0
AIDE one-stage (U3)	47.1	24.2	12.6	11	5.1	0
AIDE two-stage (U2)	31.3	23	16.7	16	12.9	0.1
AIDE two-stage (U3)	40.1	24.6	15.8	12.3	7.1	0.1
Mean patients treated						
BOIN12	4.9	5.7	6.7	7.4	5.3	
U-BOIN	6.1	6.2	6.4	6.4	5.4	
EffTox	31.8	3.8	6.3	4.7	6.1	
AIDE one-stage (U2)	4.5	4.9	6	8.3	6.3	
AIDE one-stage (U3)	4.7	5.1	6.2	8.2	5.7	
AIDE two-stage (U2)	6.1	5.4	5.5	5.8	7.3	
AIDE two-stage (U3)	6	5.5	5.5	5.8	7.3	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
BOIN12	31.4	59	8.1	0.7	0	0.8
U-BOIN	23.5	49.6	19.3	4.2	3.2	0.2
EffTox	86	10	2	0	0	1
AIDE one-stage (U2)	18.3	61	17.4	2.8	0.1	0.4
AIDE one-stage (U3)	23.2	61.2	13.2	1.9	0	0.5
AIDE two-stage (U2)	30	49.3	17.7	2.1	0	0.9
AIDE two-stage (U3)	33.4	48.9	14.7	1.9	0.1	1
Mean patients treated						
BOIN12	10.2	14	4.4	1.1	0.2	
U-BOIN	10.1	11.9	5.9	1.7	0.4	
EffTox	50.7	5.4	2.3	0.7	0.2	
AIDE one-stage (U2)	9	14.1	5.4	1.3	0.2	
AIDE one-stage (U3)	9.5	14.5	4.8	1	0.1	
AIDE two-stage (U2)	10.1	9.7	7.2	2.4	0.4	
AIDE two-stage (U3)	10.1	9.7	7.3	2.3	0.4	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
BOIN12	16.1	20.1	21.6	14.4	27.6	0.2
U-BOIN	7.1	13.1	16.3	15.4	48.1	0
EffTox	0	1	19	25	42	13
AIDE one-stage (U2)	10.6	10.8	8.3	15	55.3	0
AIDE one-stage (U3)	16.5	14.6	9.8	11.1	48	0
AIDE two-stage (U2)	9.1	8.3	9.6	13.6	58.3	1.1
AIDE two-stage (U3)	15.4	11.5	10.5	11.2	50.4	1
Mean patients treated						
BOIN12	4.3	5.2	5.8	6.6	8.1	
U-BOIN	4.2	5.5	6.8	7.1	6.7	
EffTox	5.9	3.9	11.2	11.9	22	
AIDE one-stage (U2)	3.4	4	5.1	6.8	10.6	
AIDE one-stage (U3)	3.5	4.2	5.2	6.8	10.3	
AIDE two-stage (U2)	3.7	3.9	4.6	5.5	12.2	
AIDE two-stage (U3)	3.7	3.8	4.6	5.6	12.1	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
BOIN12	20.5	56.8	12.5	0.6	0	9.6
U-BOIN	14	39.9	23	11.5	10.9	0.7
EffTox	17	41	9	2	1	30
AIDE one-stage (U2)	11	53.4	19.8	3	0.1	12.7
AIDE one-stage (U3)	13.5	55.2	16.6	2.2	0.1	12.4
AIDE two-stage (U2)	16.1	50.6	17	1.7	0	14.6
AIDE two-stage (U3)	18.8	51	14.1	1.6	0	14.5
Mean patients treated						
BOIN12	8.6	12	5.9	2	0.4	
U-BOIN	8	10	6.3	3.5	2.1	
EffTox	18.7	17.8	8	2.7	1.3	
AIDE one-stage (U2)	7.1	12.1	6.6	2.3	0.4	
AIDE one-stage (U3)	7.4	12.5	6.3	2.1	0.4	
AIDE two-stage (U2)	7.6	10.2	7	2.2	0.4	
AIDE two-stage (U3)	7.6	10.2	6.9	2.2	0.4	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
BOIN12	11.3	62.1	18.8	4.9	0.1	2.8
U-BOIN	8.5	47.5	18.6	12.4	13	0
EffTox	7	32	23	6	3	30
AIDE one-stage (U2)	10.6	45.6	27.1	12.6	1.9	2.2
AIDE one-stage (U3)	12.4	50.1	22.6	11.5	1.3	2.1
AIDE two-stage (U2)	10.5	44	28.8	9.1	1.4	6.2
AIDE two-stage (U3)	16.4	46.5	22.6	7.6	0.8	6.1
Mean patients treated						
BOIN12	6.2	10.7	7	4.1	1.7	
U-BOIN	5.9	10.8	7	4	2.4	
EffTox	11.5	14.9	13.4	4.7	3.6	
AIDE one-stage (U2)	4.6	9.9	8.1	5.2	1.9	
AIDE one-stage (U3)	4.9	10.4	7.8	4.9	1.7	
AIDE two-stage (U2)	5.1	8.6	8.2	4.7	2.3	
AIDE two-stage (U3)	5.1	8.6	8.3	4.6	2.2	

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda_T = 0.3; U3: Type = 3, Lambda_T = 1.

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
BOIN12	60.5	0.5	0	0	0	39
U-BOIN	51.8	8.3	5.7	5.7	3.7	24.8
EffTox	59	0	0	0	0	41
AIDE one-stage (U2)	58.8	1.8	0	0	0	39.4
AIDE one-stage (U3)	59	1.6	0	0	0	39.4
AIDE two-stage (U2)	61.8	1.6	0	0	0	36.6
AIDE two-stage (U3)	62.1	1.3	0	0	0	36.6
Mean patients treated						
BOIN12	20.7	2.5	0.4	0	0	
U-BOIN	17.2	4	1.5	1	0.8	
EffTox	43.1	0.5	0.1	0	0	
AIDE one-stage (U2)	20.3	2.7	0.3	0	0	
AIDE one-stage (U3)	20.4	2.6	0.2	0	0	
AIDE two-stage (U2)	21.2	2.1	0.6	0	0	
AIDE two-stage (U3)	21.1	2.1	0.6	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
BOIN12	24.4	60.8	4	0.8	0	10
U-BOIN	18.2	34.1	10.4	23.9	12.6	0.8
EffTox	9	34	7	1	0	48
AIDE one-stage (U2)	14.4	56.9	13.6	4	0	11.1
AIDE one-stage (U3)	16.5	57.2	11.8	3.2	0	11.3
AIDE two-stage (U2)	22.7	52.9	13	1.8	0	9.6
AIDE two-stage (U3)	25.8	52.2	10.8	1.4	0	9.8
Mean patients treated						
BOIN12	9.8	12.4	4.6	2	0.2	
U-BOIN	8.2	9.8	5	4.5	2.5	
EffTox	14.7	15.6	7.8	2.4	0.9	
AIDE one-stage (U2)	7.8	12.9	5.8	2.2	0.2	
AIDE one-stage (U3)	8.2	13	5.5	2.1	0.2	
AIDE two-stage (U2)	7.6	12.1	6.3	1.8	0.3	
AIDE two-stage (U3)	7.6	12.1	6.5	1.7	0.3	

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda_T = 0.3; U3: Type = 3, Lambda_T = 1.

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
BOIN12	10.7	0.7	0	0	0	88.6
U-BOIN	15.2	16.2	6	2.5	1.6	58.5
EffTox	0	0	3	1	0	95
AIDE one-stage (U2)	8.1	2	0	0	0	89.9
AIDE one-stage (U3)	8.3	1.7	0.1	0	0	89.9
AIDE two-stage (U2)	7.2	3.9	0.4	0	0	88.5
AIDE two-stage (U3)	7.3	3.7	0.4	0	0	88.6
Mean patients treated						
BOIN12	11.6	4.8	1.2	0.1	0	
U-BOIN	10.6	5.6	3.5	2.7	2.6	
EffTox	5	3.2	1.8	0.3	0	
AIDE one-stage (U2)	11.1	4.4	0.7	0.1	0	
AIDE one-stage (U3)	11.1	4.4	0.7	0.1	0	
AIDE two-stage (U2)	11.8	2.5	0.6	0	0	
AIDE two-stage (U3)	11.8	2.5	0.6	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
BOIN12	1.7	3.3	6.7	65.7	22.6	0
U-BOIN	1.5	3.2	8.2	60.1	27	0
EffTox	0	0	6	25	68	1
AIDE one-stage (U2)	2.2	2.2	3.6	54.1	37.9	0
AIDE one-stage (U3)	4	3.1	4.8	57.1	31	0
AIDE two-stage (U2)	1.2	1.1	5.8	45.4	46.4	0.1
AIDE two-stage (U3)	2.8	2.2	7.5	47.3	40.1	0.1
Mean patients treated						
BOIN12	3.2	3.4	4.7	12.9	5.8	
U-BOIN	3.5	4	5.1	9.8	7.9	
EffTox	5.3	3.3	6.7	11.4	32.9	
AIDE one-stage (U2)	3.1	3.1	3.9	12.5	7.4	
AIDE one-stage (U3)	3.1	3.2	4	13	6.7	
AIDE two-stage (U2)	3.3	3.2	3.8	8.4	11.3	
AIDE two-stage (U3)	3.3	3.2	3.8	8.4	11.3	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
BOIN12	40.7	35.7	21.7	0.4	0	1.5
U-BOIN	28.1	26.8	25.8	9.6	9.4	0.3
EffTox	57	22	10	1	0	10
AIDE one-stage (U2)	25.6	29.7	37.1	6.3	0.1	1.2
AIDE one-stage (U3)	31.9	29.3	31.9	5.6	0	1.3
AIDE two-stage (U2)	31.9	26.8	33.5	5.3	0	2.5
AIDE two-stage (U3)	37.2	27.6	28.7	4.2	0	2.3
Mean patients treated						
BOIN12	9.9	9.9	7.6	2.1	0.3	
U-BOIN	9.4	9.3	7.2	2.8	1.2	
EffTox	36.9	9.8	6.8	1.9	0.6	
AIDE one-stage (U2)	8.7	9	8.8	2.9	0.4	
AIDE one-stage (U3)	9.4	9.1	8.5	2.5	0.3	
AIDE two-stage (U2)	9.1	8.1	9.2	2.8	0.5	
AIDE two-stage (U3)	9.1	8	9.2	2.7	0.5	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
BOIN12	22.8	35.1	32.9	1	0	8.2
U-BOIN	14.4	20.7	39.5	12.9	10.9	1.6
EffTox	10	30	29	4	1	27
AIDE one-stage (U2)	18.7	29.8	32.9	7.8	0.1	10.7
AIDE one-stage (U3)	21.6	28.7	31	7.3	0.1	11.3
AIDE two-stage (U2)	19.2	24	30.4	6	0	20.4
AIDE two-stage (U3)	23.1	26.4	24.4	6	0.1	20
Mean patients treated						
BOIN12	8.5	9.4	8	2.9	0.4	
U-BOIN	7.8	7.7	8.2	4.1	2	
EffTox	15.9	13.5	13.7	4.9	1.9	
AIDE one-stage (U2)	8	8.6	8.3	3.2	0.4	
AIDE one-stage (U3)	8.3	8.7	8.2	3	0.4	
AIDE two-stage (U2)	8.9	6.5	8	2.8	0.5	
AIDE two-stage (U3)	9	6.6	7.9	2.9	0.5	

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda_T = 0.3; U3: Type = 3, Lambda_T = 1.

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
BOIN12	20.9	32.9	38	4.2	0	4
U-BOIN	8.7	20.9	38.9	16.5	14.5	0.5
EffTox	11	32	33	9	2	14
AIDE one-stage (U2)	12.9	23.4	46.3	11.8	0.7	4.9
AIDE one-stage (U3)	15.8	25.7	42.3	11	0.5	4.7
AIDE two-stage (U2)	15.9	20.3	40.9	10.7	0.7	11.5
AIDE two-stage (U3)	20.2	22.2	36.7	8.9	0.7	11.3
Mean patients treated						
BOIN12	7.2	9	9.2	3.4	0.7	
U-BOIN	6.9	8.1	8.8	4.3	1.8	
EffTox	15.7	15.1	13.9	6.9	3.4	
AIDE one-stage (U2)	6.2	8	10.3	4.1	0.9	
AIDE one-stage (U3)	6.5	8.2	10.1	3.9	0.7	
AIDE two-stage (U2)	7.1	6.3	9.4	4.1	1.4	
AIDE two-stage (U3)	7	6.3	9.5	4	1.4	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
BOIN12	12.7	51.5	5.6	0.3	0.1	29.8
U-BOIN	7.2	46	17.1	13.9	11.8	4
EffTox	10	33	5	2	0	49
AIDE one-stage (U2)	9	44.6	9.1	0.6	0	36.7
AIDE one-stage (U3)	8.9	45.4	7.9	0.1	0	37.7
AIDE two-stage (U2)	14.8	35.1	6.6	0.2	0	43.3
AIDE two-stage (U3)	15.5	35.5	5.5	0.3	0	43.2
Mean patients treated						
BOIN12	9.1	11.7	4.2	1.2	0.2	
U-BOIN	8.2	9.5	5.4	3.6	2.6	
EffTox	14.2	15.8	6	2.3	1.3	
AIDE one-stage (U2)	8.5	11.2	4.1	1.2	0.2	
AIDE one-stage (U3)	8.7	11.2	4	1.1	0.1	
AIDE two-stage (U2)	8.7	8.4	4.3	0.9	0.1	
AIDE two-stage (U3)	8.8	8.4	4.2	0.9	0.1	

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda_T = 0.3; U3: Type = 3, Lambda_T = 1.

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
BOIN12	72.7	22.9	1.2	0	0	3.2
U-BOIN	57.7	26.2	6.6	2.4	4.4	2.7
EffTox	95	1	1	0	0	4
AIDE one-stage (U2)	63.1	30	4.4	0.2	0	2.3
AIDE one-stage (U3)	67.2	27.3	3	0.2	0	2.3
AIDE two-stage (U2)	67.2	25.6	4.2	0.3	0	2.7
AIDE two-stage (U3)	69.8	23.7	3.4	0.3	0	2.8
Mean patients treated						
BOIN12	17.4	8.6	2.7	0.6	0	
U-BOIN	16.2	8.5	3.2	1	0.5	
EffTox	55.9	1.2	0.5	0.1	0	
AIDE one-stage (U2)	17.3	8.9	2.7	0.6	0	
AIDE one-stage (U3)	18.2	8.6	2.2	0.5	0	
AIDE two-stage (U2)	18	7.3	3.6	0.5	0	
AIDE two-stage (U3)	18	7.3	3.6	0.5	0.1	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
BOIN12	23.9	23.6	21.2	28.2	2.4	0.7
U-BOIN	13	15.8	16.7	37.7	16.8	0
EffTox	9	10	38	30	7	7
AIDE one-stage (U2)	17.2	12.2	12.2	43.7	14.3	0.4
AIDE one-stage (U3)	25.6	14.5	10.1	38.4	10.9	0.5
AIDE two-stage (U2)	19.2	13.8	15.1	37.1	13.1	1.7
AIDE two-stage (U3)	26.4	14.1	13.7	33.6	10.5	1.7
Mean patients treated						
BOIN12	5.3	6.1	6.6	8.6	3.3	
U-BOIN	5.8	6.8	7.1	7.2	3.4	
EffTox	15.7	6.8	15.8	11.4	7.6	
AIDE one-stage (U2)	4.3	4.8	5.9	10.4	4.6	
AIDE one-stage (U3)	4.5	5	6	10.3	4.3	
AIDE two-stage (U2)	5	4.5	5.8	8	6.4	
AIDE two-stage (U3)	5	4.5	5.8	8	6.4	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
BOIN12	3.3	41.2	34.5	19.9	1	0.1
U-BOIN	3.2	42.6	32.7	20	1.5	0
EffTox	38	16	24	20	3	0
AIDE one-stage (U2)	5.3	34.7	27	25.9	7.1	0
AIDE one-stage (U3)	6.8	41.2	26.4	21.2	4.4	0
AIDE two-stage (U2)	7.2	34.9	24.8	24.9	8	0.2
AIDE two-stage (U3)	10.2	41.6	23.7	20.5	3.8	0.2
Mean patients treated						
BOIN12	3.9	12.3	7.9	4.8	1.1	
U-BOIN	4.4	7.7	7.8	7.4	2.9	
EffTox	29.7	8.7	10.8	8.6	2.3	
AIDE one-stage (U2)	3.8	11	7.4	5.9	2	
AIDE one-stage (U3)	3.9	11.9	7.2	5.4	1.6	
AIDE two-stage (U2)	4.1	5.7	6.4	7.7	6	
AIDE two-stage (U3)	4.1	5.7	6.4	7.7	6	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
BOIN12	14.1	21.4	46.4	13.3	4.6	0.2
U-BOIN	8.5	16	45.7	17.4	12.4	0
EffTox	2	12	54	12	11	10
AIDE one-stage (U2)	13.6	14.2	34.4	25.7	12	0.1
AIDE one-stage (U3)	20	16.5	32.5	23.6	7.3	0.1
AIDE two-stage (U2)	8.7	13.4	34.7	27.2	15.1	0.9
AIDE two-stage (U3)	14.6	17.8	34.6	23.2	9	0.8
Mean patients treated						
BOIN12	4.4	5.9	9.2	6.7	3.8	
U-BOIN	4.5	6.1	9.8	6.5	3.3	
EffTox	9.9	7.8	21.4	8.9	8.1	
AIDE one-stage (U2)	3.5	4.3	8.5	8.4	5.3	
AIDE one-stage (U3)	3.6	4.5	8.8	8.4	4.7	
AIDE two-stage (U2)	3.9	4.3	7.7	7.1	6.9	
AIDE two-stage (U3)	4	4.3	7.7	7	6.9	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
BOIN12	12.3	14.7	33.2	28.1	11.4	0.3
U-BOIN	6.1	8.6	24.9	28.6	31.8	0
EffTox	0	1	43	25	26	5
AIDE one-stage (U2)	7.9	5.6	12.1	43.7	30.6	0.1
AIDE one-stage (U3)	12	7.6	11	43.6	25.7	0.1
AIDE two-stage (U2)	8.6	5.1	11.6	41.1	33.3	0.3
AIDE two-stage (U3)	12.3	6.9	13.6	40	27	0.2
Mean patients treated						
BOIN12	4.4	5.5	7.9	7.9	4.3	
U-BOIN	3.8	5	8.8	7.8	4.6	
EffTox	5.5	4	20.1	12.2	15.9	
AIDE one-stage (U2)	3.3	3.6	5.9	10.5	6.7	
AIDE one-stage (U3)	3.4	3.8	6	10.6	6.2	
AIDE two-stage (U2)	3.5	3.6	5.2	8.1	9.6	
AIDE two-stage (U3)	3.5	3.6	5.2	8.1	9.6	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
BOIN12	20.5	22.2	37.8	11	3	5.5
U-BOIN	9.1	11.4	26	26.5	27	0
EffTox	1	8	39	13	15	24
AIDE one-stage (U2)	11.5	12.3	44.5	23	5.1	3.6
AIDE one-stage (U3)	15.8	14.3	42.1	20.9	3.3	3.6
AIDE two-stage (U2)	11.9	11.7	47	19.8	5.4	4.2
AIDE two-stage (U3)	18.2	13.3	44.3	16.1	3.6	4.5
Mean patients treated						
BOIN12	6.6	7.8	8	4.9	2.4	
U-BOIN	5.9	7.5	7.8	5.3	3.6	
EffTox	6.8	8.3	17.7	7.4	10.3	
AIDE one-stage (U2)	5	6.2	9.6	6.3	2.8	
AIDE one-stage (U3)	5.2	6.3	9.7	6.1	2.4	
AIDE two-stage (U2)	5	5.5	9.9	5.4	3.4	
AIDE two-stage (U3)	5	5.5	9.8	5.4	3.5	

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
BOIN12	12.3	63.1	19.3	4.2	1.1	0
U-BOIN	10.1	61.8	19.4	5.7	3	0
EffTox	49	34	11	4	1	1
AIDE one-stage (U2)	9.2	40	31.6	15.2	3.9	0.1
AIDE one-stage (U3)	13.6	45.6	27.4	10.6	2.7	0.1
AIDE two-stage (U2)	11.6	40.2	31.4	12.3	4.3	0.2
AIDE two-stage (U3)	16.1	44	27.2	9.9	2.6	0.2
Mean patients treated						
BOIN12	5.3	13.5	7.1	3	1.1	
U-BOIN	5.8	13.2	7.5	2.7	0.8	
EffTox	34	14.6	6.9	2.5	1.4	
AIDE one-stage (U2)	4.4	11.1	8.4	4.5	1.5	
AIDE one-stage (U3)	4.6	12	8.1	4	1.3	
AIDE two-stage (U2)	4.6	8.3	8.8	5	3.3	
AIDE two-stage (U3)	4.6	8.2	8.9	5	3.2	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
BOIN12	17.6	43.5	13.5	4.9	1.9	18.6
U-BOIN	7.8	18.8	15.2	24.1	33.2	0.9
EffTox	4	15	19	10	14	38
AIDE one-stage (U2)	12.2	41.6	16.1	7.9	1.7	20.5
AIDE one-stage (U3)	16	43.1	11.6	7.6	0.8	20.9
AIDE two-stage (U2)	14.6	36.6	16.1	7	1.8	23.9
AIDE two-stage (U3)	19.1	37.8	12.6	5.9	1	23.6
Mean patients treated						
BOIN12	8.4	10.1	4.9	3.2	1.6	
U-BOIN	7.9	7.5	5.4	5	4	
EffTox	10.2	10.5	10.4	5.2	8.8	
AIDE one-stage (U2)	7.5	9.8	5.6	3.6	1.2	
AIDE one-stage (U3)	7.8	10	5.4	3.4	1.1	
AIDE two-stage (U2)	7.8	8	5.7	2.9	1.4	
AIDE two-stage (U3)	7.6	8.2	5.7	2.8	1.4	

AIDE: shared-alpha additive model; toxicity Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).

AIDE rows use IPDE alpha = 0 and continuous enrollment. U2: Type = 2, Lambda_T = 0.3; U3: Type = 3, Lambda_T = 1.

Phase I/II Operating Characteristics

Method comparison - N = 30, Cycle_Max = 2, efficacy futility cutoff = 0.85

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
BOIN12	21	49.5	10	2.2	0.2	17.1
U-BOIN	16.5	27.5	19.3	16.9	19.4	0.4
EffTox	3	16	13	3	2	63
AIDE one-stage (U2)	12.2	48.8	16.7	4.1	0.8	17.4
AIDE one-stage (U3)	16	50.2	13.5	2.6	0.5	17.2
AIDE two-stage (U2)	20.4	45.6	14.7	2.7	0.5	16.1
AIDE two-stage (U3)	25.5	44.6	11.2	2	0.4	16.3
Mean patients treated						
BOIN12	9.3	11	4.7	2.4	1	
U-BOIN	8.4	8.3	5.5	4.3	3.5	
EffTox	10.7	10.5	9.1	3.3	3	
AIDE one-stage (U2)	7.4	11.2	5.8	3	0.9	
AIDE one-stage (U3)	7.8	11.2	5.6	2.9	0.8	
AIDE two-stage (U2)	7.3	10.3	6.2	2.4	0.9	
AIDE two-stage (U3)	7.2	10.3	6.2	2.4	0.9	